# Networks Project Part 3: Congestion Control and Evaluation

Due Friday, November 20, 2026, 11:59 pm. Worth 15% of the course grade. Work alone or in pairs.

## Overview

In Parts 1 and 2 you built a reliable transport protocol over UDP with Go-Back-N and then Selective Repeat. In Part 3 you add congestion control and measure your protocol against a provided TCP baseline on an emulated link.

## Requirements

- Implement slow start and AIMD congestion avoidance with a congestion window measured in segments.
- Implement fast retransmit on three duplicate acknowledgements and fast recovery as in TCP Reno.
- Run your protocol on the provided emulator at 10 Mbit/s with 50 ms RTT and loss rates of 0%, 1% and 5%.
- Plot cwnd over time for each loss rate and report average throughput for each run.
- Your code should be well tested.
- Submit a 3-page report explaining your design choices and the plots.

## Grading

- Correct slow start and AIMD: 35 points
- Fast retransmit and fast recovery: 25 points
- Measurements and plots: 25 points
- Report: 15 points

Question 3 of the report asks: with a 5% loss rate, what is the expected steady-state throughput of your protocol with MSS = 1460 bytes and RTT = 50 ms, using the loss-rate throughput formula from Module 4? Show your work.